Discuss 10 emerging mobile computing technologies highlighting there application advantages and disadvantages and how thay are transforming mobile computing

Mobile computing is being transformed by a stack of emerging technologies that make phones more intelligent (AI), more immersive (XR), more connected (5G-Advanced/6G/Wi-Fi 7/satellite), more flexible (foldables), and more secure (passkeys), while pushing heavy computation closer to users (edge). The result is lower latency, better privacy, new device form factors, and new application categories that were impractical on earlier mobile platforms.[1] [2]

10 emerging technologies

Emerging technology	Applications (examples)	Advantages	Disadvantages	How it transforms mobile computing
On-device AI (edge AI)	Real-time translation, camera enhancement, voice assistants running locally [3] [4]	Lower latency and improved privacy because data can stay on the device [3] [4]	Higher device cost/power use and model size limits vs cloud AI [3]	Shifts smartphones from “app launchers” to always-available intelligent agents that can work offline [3] [4]
Generative AI on mobile	Creating text/images, summarizing content, generating UI assets inside apps [5] [6]	Personalization and automation of tasks that used to require PCs or cloud-only tools [5]	Risk of hallucinations and increased compute/battery demand [5]	Moves content creation and decision support into the phone, changing mobile apps into assistant-like workflows [5] [6]
Mobile edge computing (MEC)	Cloud gaming, AR rendering, real-time analytics for smart-city/mobile apps [7] [8]	Very low latency and reduced backhaul congestion by processing closer to users [7] [8]	Deployment complexity and uneven availability by region/operator [7]	Extends the “compute capacity” of phones by offloading heavy workloads to nearby edge servers [7] [8]

Emerging technology	Applications (examples)	Advantages	Disadvantages	How it transforms mobile computing
Private 5G + edge	Factory mobility, hospital devices, logistics tracking with local control [1] [9]	Dedicated performance, better control/security, and customized enterprise use cases [1] [9]	Higher cost and operational complexity for organizations [1] [9]	Expands mobile computing beyond consumers into mission-critical enterprise systems with predictable connectivity [1] [9]
5G-Advanced (3GPP Release 18)	Enhanced XR/industrial IoT performance and smarter network behavior [10] [11]	More intelligent and capable 5G evolution (e.g., tighter XR support and AI/ML integration themes) [10] [11]	Requires new network/device upgrades and rollout time [10]	Improves reliability/latency and makes immersive and industrial mobile applications more practical at scale [10] [11]
6G (IMT-2030 direction)	Integrated sensing/communication, advanced AI-enabled applications, expanded coverage goals [2]	Targets new capabilities such as sensing-related functions and AI-related capabilities across the system [2]	Still early; standards, spectrum, and device ecosystem are not mature yet [2]	Redefines “connectivity” into a context-aware platform (communications + sensing + AI execution) for future mobile services [2]
Wi-Fi 7	High-throughput AR/VR and smoother video/data-heavy mobile use [12]	Higher speeds, reduced latency, and efficiency features that can improve device battery life in some scenarios [12]	Requires new routers/clients; benefits depend on local network quality [12]	Makes local wireless networks fast enough to behave like “short-range fiber,” enabling richer in-home/office mobile experiences [12]
XR on mobile (AR/VR/MR)	Training, remote assistance, immersive collaboration and entertainment [5] [13]	More immersive interaction models and new app categories (spatial computing) [13] [5]	Hardware cost/comfort constraints and motion/latency sensitivity [14] [13]	Shifts interfaces from flat touchscreens to spatial, sensor-driven computing experiences [13] [14]

Emerging technology	Applications (examples)	Advantages	Disadvantages	How it transforms mobile computing
Foldable / multi-screen devices	Tablet-like productivity, multitasking, large-screen reading/design on a pocket device [15] [16]	Bigger usable display with portability; enables new UI layouts and multitasking patterns [16]	Durability concerns and higher prices than slab phones (market still evolving) [15] [16]	Turns the phone into an adaptive form factor (phone ↔ tablet), pushing developers toward responsive, multi-pane mobile UIs [16] [15]
Passkeys (passwordless authentication)	Logging into apps/websites with phishing-resistant cryptographic credentials [17]	Faster sign-in and phishing-resistant authentication using cryptographic key pairs [17]	Account recovery and cross-device workflows can still confuse users depending on ecosystem setup [17]	Improves trust and reduces credential theft, enabling safer mobile payments, identity, and enterprise access from phones [17]

What to focus on in practice

- For app developers: prioritize on-device AI for low-latency/privacy features, and design for foldables and XR form factors where relevant.[16] [3]
- For networked apps (streaming, telemedicine, gaming): design for edge + 5G/5G-Advanced to reduce latency and jitter.[10] [7]
- For security-sensitive systems: adopt passkeys to cut phishing and password reuse risks in mobile-first access.[17]

Which area is most important for the assignment context—consumer apps, enterprise/industry, or telecom/network evolution?

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